

AP CALCULUS AB – UNIT 2

Differentiation: Definition and Fundamental Properties

Comprehensive Coloured Review



Name: _____ Class: _____ Date: _____

Review Scope and Learning Objectives

- Compute and interpret average and instantaneous rates of change.
- Use the limit definition of the derivative and write tangent-line equations.
- Estimate derivatives and analyze differentiability and continuity from multiple representations.
- Differentiate algebraic, exponential, logarithmic, and trigonometric functions using foundational, product, and quotient rules.

Website Lesson Map

Lesson	Website title	Review target
2.1	Rates of Change and Tangent Lines	Connect definitions, derivative values, and symbolic rules.
2.2	Tangent Lines and the Derivative	Connect definitions, derivative values, and symbolic rules.
2.3	The Derivative: Graphs, Differentiability, and Continuity	Connect definitions, derivative values, and symbolic rules.
2.4	Basic Derivative Rules	Connect definitions, derivative values, and symbolic rules.
2.5	Product and Quotient Rules	Connect definitions, derivative values, and symbolic rules.

Key Knowledge, Formulas, and Theorems

- Average rate: $\frac{f(b) - f(a)}{b - a}$; instantaneous rate: $f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$.
- Tangent line at $x = a$: $y - f(a) = f'(a)(x - a)$.
- Differentiability implies continuity; continuity alone does not imply differentiability.
- Power rule: $\frac{d}{dx}x^n = nx^{n-1}$; $\frac{d}{dx}e^x = e^x$; $\frac{d}{dx}\ln x = 1/x$.
- Product: $(fg)' = f'g + fg'$; quotient: $(f/g)' = \frac{f'g - fg'}{g^2}$.

Strategy and Precision Checklist

1. Identify the representation and the exact quantity requested before calculating.
2. Write the relevant definition, theorem, or derivative rule before substitution.
3. Preserve exact values in non-calculator work; use radian mode and appropriate rounding in calculator work.
4. For contextual conclusions, include the input, sign, units, and meaning.
5. Justifications must state why a theorem or procedure applies, not merely name it.

Solved Mixed Examples

Solved Example: Example 1 – 2.2

Which limit represents $f'(2)$?

Answer and Reasoning

Result: $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$. A derivative at a point is the limit of the difference quotient over an increment tending to zero.

Solved Example: Example 2 – 2.2

Find the tangent line to $y = x^2 - 2x + 3$ at $x = -2$.

Answer and Reasoning

Result: $y = -6x - 1$. Compute the point $(a, f(a))$ and slope $f'(a)$, then use point-slope form.

Solved Example: Example 3 – 2.3

A table gives $f(\frac{39}{10}) = -\frac{68}{5}$ and $f(\frac{41}{10}) = -\frac{72}{5}$. Estimate $f'(4)$ using a symmetric difference.

Answer and Reasoning

Result: -4 . Use $[f(c+h) - f(c-h)]/(2h)$.

Solved Example: Example 4 – 2.3

Which condition is sufficient to conclude that f is continuous at $x = c$?

Answer and Reasoning

Result: f is differentiable at $x = c$. Differentiability implies continuity. The other statements alone are insufficient.

Leveled Mixed Practice

Foundation

Questions in this band integrate the website lessons at an increasingly demanding level.

1. For $f(x) = x^2 - 4x$, find the average rate of change on $[4, 5]$.

2. If $f(x) = 2x^2 - 1x + 1$, evaluate $\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$.

3. Which limit represents $f'(2)$?

4. Find the tangent line to $y = x^2 - 2x + 3$ at $x = -2$.

Core

Questions in this band integrate the website lessons at an increasingly demanding level.

5. A table gives $f(\frac{39}{10}) = -\frac{68}{5}$ and $f(\frac{41}{10}) = -\frac{72}{5}$. Estimate $f'(4)$ using a symmetric difference.

6. Which condition is sufficient to conclude that f is continuous at $x = c$?

7. Differentiate $f(x) = 3x^5 + 2x^2 - 3$.

8. Find $\frac{d}{dx}[\log(x)]$.

Advanced

Questions in this band integrate the website lessons at an increasingly demanding level.

9. Differentiate $(x^2 + 1)(2x - 1)$.

10. Differentiate $\frac{x^2 + 1}{x + 2}$.

11. At $x = 2$, $f = 4$, $g = 3$, $f' = 1$, and $g' = 3$. Find $(fg)'(2)$.

12. At $x = 1$, $f = 4$, $g = 3$, $f' = 1$, and $g' = -3$. Find $(f/g)'(1)$.

Difficult

Questions in this band integrate the website lessons at an increasingly demanding level.

13. For $f(x) = 3x^2 + x$, find the average rate of change on $[1, 2]$.

14. If $f(x) = 3x^2 + 3x + 1$, evaluate $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$.

15. A tangent to $y = \ln x$ is drawn at $x = 1.5$. What is its slope to three decimal places?

16. A calculator table gives $f(2.2) = 4.1$, $g(2.2) = 4.7$, $f'(2.2) = 2.6$, and $g'(2.2) = 2.8$. Find $(fg)'(2.2)$.

Challenge

Questions in this band integrate the website lessons at an increasingly demanding level.

17. A population is $P(t) = 4.2e^t$ thousand. What is the instantaneous growth rate at $t = 2.46$?

18. Use a calculator to evaluate the instantaneous rate of change of $f(x) = 3.35x^2 + 0.10x$ at $x = 1.87$.

19. Use a calculator to find $f'(0.76)$ for $f(x) = \sin x + e^x$.

20. A tangent to $y = \ln x$ is drawn at $x = 2.37$. What is its slope to three decimal places?

AP-Style Mixed Checkpoint

1. Differentiate $(x^2 + 3)(4x - 1)$.

(A) $12x^2 - 2x + 12$

(B) $8x$

(C) $4x^2 + 12$

(D) $2x(4x - 1)$

2. Differentiate $\frac{x^2 + 4}{x + 3}$.

(A) $\frac{x^2 + 2x(x+3) + 4}{(x+3)^2}$

(B) $2x$

(C) $\frac{2x}{x+3}$

(D) $\frac{-x^2 + 2x(x+3) - 4}{(x+3)^2}$

3. At $x = 3$, $f = 1$, $g = 2$, $f' = 1$, and $g' = 1$. Find $(fg)'(3)$.

(A) 4

(B) -3

(C) 3

(D) 2

4. At $x = 1$, $f = 4$, $g = 1$, $f' = -2$, and $g' = 4$. Find $(f/g)'(1)$.

(A) -19

(B) 18

(C) -17

(D) -18

5. For $f(x) = x^2 - 3x$, find the average rate of change on $[2, 3]$.

(A) 1

(B) -2

(C) 2

(D) 3

6. If $f(x) = 2x^2 - 3x + 1$, evaluate $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$.

(A) 0

(B) -1

(C) 1

(D) 2

7. Which limit represents $f'(1)$?

(A) $\lim_{x \rightarrow 1} f(x)$

(C) $\lim_{h \rightarrow 0} \frac{f(1) - f(h)}{1 - h}$

(B) $\lim_{h \rightarrow 1} \frac{f(h) - f(1)}{h}$

(D) $\lim_{h \rightarrow 0} \frac{f(1 + h) - f(1)}{h}$

8. Find the tangent line to $y = 2x^2 - 2x + 1$ at $x = 1$.

(A) $y = 2x$

(B) $y = 3x - 2$

(C) $y = 2x + 3$

(D) $y = 2x - 1$

Free-Response Synthesis

FRQ1 – Rate of change and tangent line

Let $f(x) = x^3 - 4x + 2$.

(a) Write $f'(2)$ as a limit of a difference quotient.

[2 points]

(b) Find $f'(2)$ exactly.

[2 points]

(c) Write the tangent-line equation at $x = 2$.

[2 points]

FRQ2 – Derivative values from a table

Differentiable functions f and g have $f(1) = 3$, $g(1) = 2$, $f'(1) = -1$, and $g'(1) = 4$.

(a) Find $(fg)'(1)$.

[2 points]

(b) Find $(f/g)'(1)$.

[3 points]

(c) Interpret $f'(1) = -1$ if f is measured in liters and x in minutes.

[2 points]

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